

Dr. Shibiao WAN

Assistant Professor

Co-Director for Bioinformatics and Systems Biology (BISB) PhD Program

University of Nebraska Medical Center

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Highlights

- **>15 years of experience** in artificial intelligence, machine learning, bioinformatics, and computational biology
- **>70 publications**
- **Google Scholar Citations: >2000**
- **Editor-in-Chief** for *Current Proteomics* (Elsevier)
- **Associate Editor** for *Heliyon* (Cell Press)
- **Editorial Board Member** for *Briefings in Functional Genomics* (Oxford University Press)
- **Editorial Board Member** for *BMC Bioinformatics* (Springer Nature)
- **Editorial Board Member** for *Discover Oncology* (Springer Nature)
- **Editorial Board Member** for *PeerJ Computer Science* (PeerJ)
- **Editorial Board Member** for *International Journal of Microbiology* (Wiley)
- **Editorial Board Member** for *Computational and Mathematical Methods* (Wiley)
- **Guest Associate Editor** for (1) *Frontiers in Immunology*; (2) *Frontiers in Cell and Developmental Biology*; (3) *Frontiers in Pharmacology*; (4) *Frontiers in Oncology*; (5) *Frontiers in Genetics*; (6) *Frontiers in Psychology*; (7) *International Journal of Molecular Sciences*; (8) *Biology*; (9) *Genes*; (10) *BMC Methods*; (11) *Discover Life*.
- **Session Chair** for *The 2026 International Conference on Intelligent Biology and Medicine (ICIBM 2026)*
- **Co-Chair** for *Publicity Committee* for *ICIBM 2025*
- **Scientific Program Committee (SPC) Member** for *AMIA 2025 Annual Symposium*
- **Technical Program Committee (TPC) Member** for >20 AI/ML international conferences including *IEEE ICTAI*
- **Reviewer** for >50 journals including *Nature Biotechnology*; *Nature Methods*; *Nature Communications*; *Nature Computational Science*; *Science Advances*; *Advanced Science*; *Cancer Research*; *Nucleic Acids Research*.
- **Top Reviewers Award** (top 1%) in *Cross-Field*, and *Biology and Biochemistry* awarded by Clarivate
- Developed >20 highly accessed bioinformatics tools/webservers
- **Springer Nature Editor of Distinction Award** by Springer Nature (2025)
- **New Investigator Award** by University of Nebraska Medical Center (2024)
- **FIRST Award** by Nebraska EPSCoR Program (2023)
- **Outstanding Young Alumni Award** by The Hong Kong Polytechnic University (2022)
- **IEEE Senior Member**

Curriculum Vitae

ACADEMIC EXPERIENCE

- 2025 - present *Co-Director for Bioinformatics and Systems Biology (BISB) PhD Program*, [University of Nebraska Medical Center](#), Omaha, NE, USA
- 2024 - present *Assistant Professor (courtesy)*, Department of Biostatistics, [University of Nebraska Medical Center](#), Omaha, NE, USA
- 2022 - present *Assistant Professor*, Department of Genetics, Cell Biology and Anatomy, [University of Nebraska Medical Center](#), Omaha, NE, USA
- 2022 - 2025 *Assistant Director for Bioinformatics and Systems Biology Core*, [University of Nebraska Medical Center](#), Omaha, NE, USA
- 2019 - 2022 *Bioinformatics Research Scientist*, [St. Jude Children's Research Hospital](#), Memphis, TN, USA
- 2017 – 2019 *Postdoctoral Researcher*, [University of Pennsylvania](#), Philadelphia, PA, USA
- 2016 – 2017 *Postdoctoral Research Associate*, [Princeton University](#), Princeton, NJ, USA
- 2014 – 2016 *Postdoctoral Fellow*, [The Hong Kong Polytechnic University](#), Hong Kong SAR
- 2014 *Research Associate*, [The Hong Kong Polytechnic University](#), Hong Kong SAR
- 2013 *Visiting Scholar*, [Johns Hopkins University School of Medicine](#), MD, USA and [CBIL lab](#) of [Virginia Tech](#), VA, USA.

EDUCATION

- 2010 – 2014 *Ph.D.*, [The Hong Kong Polytechnic University](#), Hong Kong SAR
Dissertation Advisor: Dr. Man-Wai Mak
- 2006 – 2010 *B.Eng.*, [Wuhan University](#), Wuhan, China.
Thesis Advisor: Prof. Deshi Li

RESEARCH INTERESTS

- **Bioinformatics and Computational Biology:**
 - Single-Cell Analysis, Spatial Omics Analysis, Multi-Omics Analysis, Alternative Splicing.
 - Cancer Research, Cancer Disparities, Leukemia, Medulloblastoma, Breast Cancer, Lung Cancer.
 - NGS Data Analysis, Genomics, Epigenomics, Proteomics.
 - Intelligent Healthcare, Precision Medicine, Alzheimer's Disease, Parkinson's Disease.
 - Sequence Analysis, Antimicrobial Peptide Identification, Protein Subcellular Localization.
- **Artificial Intelligence (AI) and Machine Learning (ML):**
 - Deep Learning, Large Language Models, Transfer Learning, Interpretable Machine Learning.
 - Big Data Analytics, High-Dimensional Data Analysis, Dimension Reduction.
 - Unsupervised Learning, Supervised Learning, Data and Text Mining, Computational Modeling.
 - Semi-Supervised Learning, Transductive Learning, Kernel Learning, Ensemble Learning.
 - Multi-Label Classification, Discriminative Models, Regression Analysis, Optimization.

RESEARCH FUNDING

➤ Active Grants:

- **Number:** 2500836 (Wan)
Role: PI
Sponsor: NSF/Smart and Connected Health
Title: SCH: A Multi-Modal Transfer Learning Framework to Maximize Health Outcomes for Breast Cancer
Dates: 08/01/2025 – 07/31/2029
Effort: 2 calendar months
Amount: \$1,000,000 (Dr. Wan's portion: \$1,000,000)
- **Number:** 1R03OD038391-01 (Wan)
Role: PI
Sponsor: DHHS/NIH/Office of The Director
Title: Leveraging Heterogeneous Common Fund Data Sets and Beyond for Identifying Lung Cancer Subtypes
Dates: 09/05/2024 – 09/04/2026
Effort: 3.6 calendar months
Amount: \$307,000 (Dr. Wan's portion: \$307,000)
- **Number:** 3P20GM103427-23S2 (Sorgen)
Role: co-Project Leader
Sponsor: DHHS/NIH/NIGMS
Title: Nebraska Research Network in Functional Genomics
Dates: 06/20/2024 – 04/30/2026
Effort: 3 calendar months
Amount: \$958,437 (Dr. Wan's portion: \$216,668)
- **Number:** 31-7210-6605 (Wan)
Role: PI
Sponsor: UNMC Child Health Research Institute (CHRI)
Title: An Integrated Multi-Modal Artificial Intelligence Approach for Categorizing Pediatric Acute Myeloid Leukemia
Dates: 07/01/2024 – 06/30/2026
Effort: 0.6 calendar months
Amount: \$50,000 (Dr. Wan's portion: \$50,000)

Curriculum Vitae

- **Number:** PR210736 (Wang)
Role: co-Investigator
Sponsor: DoD/CMDRP
Title: The Development of a Macromolecular Dexamethasone Prodrug for the Treatment of Focal Segmental Glomerulosclerosis (FSGS)
Dates: 04/01/2022 – 03/31/2026
Effort: 0.6 calendar month
Amount: \$2,449,816 (Dr. Wan's portion: 5% effort and a PhD student's salary)
- **Number:** 3P20GM152326-02S1 (Gundry)
Role: co-Investigator
Sponsor: DHHS/NIH/NIGMS
Title: Nebraska Center for Heart and Vascular Research Administrative Supplements for Research on Women's Health
Dates: 01/01/2025 – 12/31/2026
Effort: 1.2 calendar month
Amount: \$307,000 (Dr. Wan's portion: 10% effort)
- **Completed Grants:**
 - **Number:** 3P30CA036727-37S5 (Sweasy)
Role: Project Leader/Point of Contact (POC)
Sponsor: DHHS/NIH/NCI
Title: An Accurate Machine Learning Framework for Childhood Acute Myeloid Leukemia Subtype Identification by Integrating Bulk and Single-Cell Multi-Omics Data Within and Beyond the CCDI System
Dates: 09/26/2023 – 08/31/2025
Effort: 3.0 calendar month
Amount: \$500,000 (Dr. Wan's portion: \$500,000)
 - **Number:** 31-3205-1243 (Wan)
Role: PI
Sponsor: Nebraska Research Initiative (NRI) Collaborative Grant
Title: An Artificial Intelligence Based Framework to Reduce Health Disparities for Breast Cancer
Dates: 07/01/2024 – 06/30/2025
Effort: 0.6 calendar month
Amount: \$100,000 (Dr. Wan's portion: \$100,000)

Curriculum Vitae

- **Number:** IRG-22-146-07-IRG (Wan)

Role: PI

Sponsor: American Cancer Society Institutional Research Grant (BCC-ACS-IRG) Seed Grants Programs

Title: Tackling Racial Disparities for Prostate Cancer by a Multi-Modal Transfer Learning Framework

Dates: 02/01/2024 – 09/12/2025

Effort: 0.6 calendar month

Amount: \$60,000 (Dr. Wan's portion: \$60,000)

- **Number:** 1P50AA030407-01 Pilot Project (Wan)

Role: PI

Sponsor: Alcohol Center of Research-Nebraska (ACORN) Pilot Projects Program

Title: Machine Learning for Identifying Biomarkers Within and Between Brain Networks for Alcohol Use Disorder Diagnosis

Dates: 07/01/2023 – 01/31/2025

Effort: 0.6 calendar month

Amount: \$25,000 (Dr. Wan's portion: \$25,000)

- **Number:** 31-7210-6586 (Wan)

Role: PI

Sponsor: BCC-CHRI-PCRG Pilot Projects Program

Title: Advanced Machine Learning for Identifying Subtypes of B-Cell Acute Lymphoblastic Leukemia Based on Transcriptomic Profiling

Dates: 02/01/2023 – 01/31/2025

Effort: 1.0 calendar month

Amount: \$50,000 (Dr. Wan's portion: \$50,000)

- **Number:** OIA-2044049 (Wan)

Role: PI

Sponsor: Nebraska EPSCoR FIRST Award

Title: A Unified Machine Learning Framework to Integrate Large-Scale Single Cell and Bulk Multi-Omics Data for Cancer Subtyping

Dates: 12/01/2023 – 11/30/2024

Effort: 0.6 calendar month

Amount: \$25,000 (Dr. Wan's portion: \$25,000)

Curriculum Vitae

- **Number:** 3P20GM130447-04S1 (Dunaevsky)

Role: co-I

Sponsor: DHHS/NIH/NIGMS

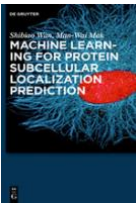
Title: Cognitive Neuroscience of Development and Aging Center Team Science Administrative Supplement

Dates: 08/01/2023 – 01/31/2025

Effort: 1.2 calendar month

Amount: \$939,079 (Dr. Wan's portion: \$36,728)

PUBLICATIONS



BOOK

1. **S. Wan** and M. W. Mak, “[Machine Learning for Protein Subcellular Localization Prediction](#)”, ISBN 978-1-5015-0150-0, published by **De Gruyter**, Germany, 2015.
2. **S. Wan**, Y. Fan, C. Jiang and S. Li, “[Bioinformatics and Machine Learning for Cancer Biology](#)”, published by **MDPI**, ISBN 978-3-0365-4814-2, Switzerland, 2022. (Edited Book)

JOURNALS (Impact Factor (IF) from Journal Citation Reports (JCR); *: corresponding author; #: co-first author)

3. M. Sun, J. Wang, and **S. Wan***, “RaMBat: Accurate Identification of Medulloblastoma Subtypes from Diverse Data Sources with Severe Batch Effects”, **Molecular Oncology**, 2026. (**JCR Q1; IF: 4.5**)
4. N. Peterson, M. Sun, J. Wang and **S. Wan***, “MetaPaCS: A Novel Meta-Learning Framework for Pancreatic Cancer Subtype Identification”, **bioRxiv**, 2025, 2025.12.29.696875.
5. M. J. Baek, L. Li, J. Wang, V. Band and **S. Wan***, “MOTLAB: A Weighted Multi-Omics Transfer Learning Approach to Mitigate Breast Cancer Racial Disparities”, **bioRxiv**, 2025, 2025.12.22.696111.
6. L. Li, J. Wang, and **S. Wan***, “A Multi-Modal Transfer Learning Framework to Reduce Health Disparities in Prostate Adenocarcinoma”, **bioRxiv**, 2025, 2025.12.15.694538.
7. Y. Ma[#], X. Wu[#], R. Zhu[#], Q. Sun[#], F. Yang[#], Z. Wang[#], S. Dai[#], X. Qiu, X. Yu, W. Sun, K. Chen, H. Wang, L. Cheng, S. Xie, T. Xu, Y. Chen, W. Ji^{*}, **S. Wan***, and J. Wang^{*}, “Mapping Structural and Molecular Spatiotemporal Dynamics of Macaque Brain During Development and Aging”, **Research Square**, 2025.
8. L. Li, H. Xiao, X. Wu, Z. Tang, J. Khoury, J. Wang, and **S. Wan***, “RanBALL: An Ensemble Machine Learning Framework for Accurate Subtype Identification of Pediatric B-Cell Acute Lymphoblastic Leukemia”, **Advanced Intelligent Systems**, 2025, e202500965. (**JCR Q1; IF: 6.1**)
9. T. J. Huggins, D. J. Chai, **S. Wan** and C. F. Lei, “Editorial: Subjective well-being and human decision behaviors”, **Frontiers in Psychology**, 2025, 16:1722414. (**JCR Q1; IF: 2.9**)

10. X. Wu, J. Wang, and **S. Wan***, “RPSLearner: A Novel Approach Based on Random Projection and Deep Stacking Learning for Categorizing NSCLC”, *Advanced Intelligent Systems*, 2025, e202500635. (**JCR Q1; IF: 6.1**)
11. B. Kennedy, M. Raza, S. Mirza, A. Rajan, F. Oruji, M. Storck, S. Lele, T. Reznicek, L. Li, J. Rowley, **S. Wan**, B. Mohapatra*, H. Band*, and V. Band*, “ECD Co-Operates with ERBB2 to Promote Tumorigenesis through Upregulation of Unfolded Protein Response and Glycolysis”, *Cancer Letters*, 2025, vol. 632, pp. 217959. (**JCR Q1; IF: 10.1**)
12. L. Li, J. Khoury, J. Wang, and **S. Wan***, “AttentionAML: An Attention-Based Deep Learning Framework for Accurate Molecular Categorization of Acute Myeloid Leukemia”, *bioRxiv*, 2025, 2025.05.20.655179.
13. C. Zhang, X. Zhu, N. Peterson, J. Wang, and **S. Wan***, “A Comprehensive Review on RNA Subcellular Localization Prediction”, 2025, *arXiv*, arXiv:2504.17162.
14. J. Fang, S. Singh, B. Wells, Q. Wu, H. Jin, L. Janke, **S. Wan**, J. Steele, J. P. Connelly, A. Murphy, R. Wang, A. M. Davidoff, M. Ashcroft, S. M. Pruett-Miller, and J. Yang, “The Context-Dependent Epigenetic and Organogenesis Programs Determine 3D vs. 2D Cellular Fitness”, *eLife*, 2025, vol. 14, pp. RP101299. (**JCR Q1; IF: 6.4**)
15. G. Feng, H. Xu, **S. Wan**, H. Wang, X. Chen, R. Magan, Y. Han, Y. Wei, and H. Gu, “Twelve Practical Recommendations for Developing and Applying Clinical Predictive Models”, *The Innovation Medicine*, 2024, vol. 2, no. 4, pp. 100105.
16. J. Feng[#], M. Sun[#], C. Liu, W. Zhang, C. Xu, J. Wang, G. Wang, and **S. Wan***, “SAMP: Identifying Antimicrobial Peptides by an Ensemble Learning Model Based on Proportionalized Split Amino Acid Composition”, *Briefings in Functional Genomics*, 2024, vol. 23, no. 6, pp. 879-890.
17. Z. Xu, L. Li, R. Liu, M. Azzam, **S. Wan**, and J. Wang, “Functional Connectivity Alterations in Cocaine Use Disorder: Insights from the Triple Network Model and the Addictions Neuroclinical Assessment Framework”, *bioRxiv*, 2024, 2024.11.12.623073.
18. M. Azzam, Z. Xu, R. Liu, L. Li, K. M. Soh, K. B. Challagundla, **S. Wan**, and J. Wang*, “A review of artificial intelligence-based brain age estimation and its applications for related diseases”, *Briefings in Functional Genomics*, 2024, elae042.
19. H. Xiao, J. Wang*, and **S. Wan***, “WIMOAD: Weighted Integration of Multi-Omics Data for Alzheimer's Disease (AD) Diagnosis”, *bioRxiv*, 2024, 2024.09.25.614862.
20. H. Chen, Y. Xu, H. Lin, **S. Wan***, and L. Luo*, “A Prognostic Framework for Predicting Lung Signet Ring Cell Carcinoma via a Machine Learning Based Cox Proportional Hazard Model”, *Journal of Cancer Research and Clinical Oncology*, 2024, vol. 150, no. 364, pp. 1-15. (**JCR Q2; Rank: 79/214; IF: 2.7**)
21. L. Li[#], M. Sun[#], J. Wang, and **S. Wan***, "Multi-Omics Based Artificial Intelligence for Cancer Research", *Advances in Cancer Research*, 2024, vol. 163, pp. 303-356.
22. Z. Ahmed[#], **S. Wan**[#], F. Zhang[#], and W. Zhong[#], “Artificial Intelligence for Omics Data Analysis”, *BMC Methods*, 2024, vol. 1, no. 1, 4.

23. H. Xiao, Y. Zou, J. Wang, and S. Wan*, "A Review for Artificial Intelligence Based Protein Subcellular Localization", *Biomolecules*, 2024, vol. 14, no. 4, 409. (*JCR Q1; Rank: 70/285; IF: 5.5*)
24. J. Zeng, Y. Weng, T. Lai, L. Chen, Y. Li, Q. Huang, S. Zhong, S. Wan*, and L. Luo*, "Procyanidin alleviates ferroptosis and inflammation of LPS-induced RAW264.7 cell via the Nrf2/HO-1 pathway", *Naunyn-Schmiedeberg's Archives of Pharmacology*, 2024, vol. 397, no. 6, pp. 4055-4067. (*JCR Q2; Rank: 114/278; IF: 3.6*)
25. M. Sun[#], L. Li[#], H. Xiao[#], J. Feng[#], J. Wang, and S. Wan*, "Editorial: Bioinformatics Analysis of Omics Data for Biomarker Identification in Clinical Research, Volume II", *Frontiers in Genetics*, 2023, vol. 14, 1256468. (*JCR Q2; Rank: 48/176; IF: 4.599*)
26. H. J. Mallard, S. Wan, P. Nidhi, Y. D. Hanscom-Trofy, B. Mohapatra, N. T Woods, J. A. Lopez-Guerrero, A. Llombart-Bosch, I. Machado, K. Scotlandi, N. F. Kreiling, M. C. Perry, S. Mirza, D. W. Coulter, V. Band, H. Band, and G. Ghosal, "USP1 Expression Driven by EWS::FLI1 Transcription Factor Stabilizes Survivin and Mitigates Replication Stress in Ewing Sarcoma", *Molecular Cancer Research*, 2023, vol. 21, no. 11, pp. 1186-1204. (*JCR Q2; Rank: 72/241; IF: 5.2*)
27. Y. Liang, Z. Su, X. Mao, S. Wan* and L. Luo*, "Editorial: Ferroptosis as a Novel Therapeutic Target for Inflammation-Related Diseases", *Frontiers in Pharmacology*, 2023, vol. 14, article 1152326. (*JCR Q1; Rank: 50/279; IF: 5.988*)
28. W. Shi, S. Mirza, M. Kuss, B. Liu, A. Hartin, S. Wan, Y. Kong, B. Mohapatra, H. Band, V. Band and B. Duan, "Embedded Bioprinting of Breast Tumor Cells and Organoids Using Low Concentration Collagen based Bioinks", *Advanced Healthcare Materials*, 2023, 2300905. (*JCR Q1; Rank: 8/96; IF: 10.0*)
29. J. Wang and S. Wan*, "Editorial: Single Cell Meets Metabolism and Cancer Biology", *Frontiers in Oncology*, 2023, vol. 13, article 1125186. (*JCR Q2; Rank: 78/245; IF: 5.738*)
30. Y. Liu, J. Klein, R. Bajpai, Q. Tran, P. Kolekar, J. L. Smith, R. E. Ries, L. Dong, B. J. Huang, J. Wang, T. Alonzo, L. Tian, H. L. Mulder, K. Szlachta, T. I. Shaw, J. Ma, M. Walsh, G. Song, T. Westover, R. Autry, A. Gout, D. Wheeler, S. Wan, G. Wu, J. J. Yang, W. Evans, M. Loh, J. Easton, J. M. Klco, S. Meshinchi, P. A. Brown, S. M. Pruett-Miller and X. Ma, "Etiology of Oncogenic Fusions in 5,190 Childhood Cancers and Its Clinical and Therapeutic Implication", *Nature Communications*, 2023, vol. 14, no. 1739, pp. 1-18. (*JCR Q1; Rank: 4/72; IF: 14.919*)
31. S. Wan*, C. Jiang, S. Li and Y. Fan, "Special Issue on Bioinformatics and Machine Learning for Cancer Biology", *Biology*, 2022, vol. 11, no. 3, 361. (*JCR Q1; Rank: 16/93; IF: 5.079*)
32. T. Sakamoto, K. Batmanov, S. Wan, Y. Guo, L. Lai, R. B. Vega and D. P. Kelly, "The Nuclear Receptor ERR Cooperates with the Cardiogenic Factor GATA4 to Orchestrate Transcriptional Control of Cardiomyocyte Differentiation", *Nature Communications*, 2022, vol. 13, no. 1991, pp. 1-20. (*JCR Q1; Rank: 4/72; IF: 14.919*)
33. S. Wan* and J. Wang*, "A Sequence Obfuscation Method for Protecting Personal Genomic Privacy", *Frontiers in Genetics*, 2022, vol. 13, article 876686. (*JCR Q2; Rank: 48/176; IF: 4.599*)
34. R. Wang, X. Zheng, J. Wang, S. Wan, M. H. Wong, K. S. Leung and L. Cheng, "Improving Bulk RNA-seq Classification by Transferring Gene Signature from Single Cells in Acute Myeloid Leukemia", *Briefings in Bioinformatics*, 2022, vol. 23, no. 2, bbac002. (*JCR Q1; Rank: 3/78; IF: 11.622*)

35. V. Honnell, J. Norrie, A. Patel, C. Ramirez, J. Zhang, K. Lai, **S. Wan** and M. A. Dyer, "Identification of a Modular Super-Enhancer in Murine Retinal Development", *Nature Communications*, 2022, vol. 13, no. 253, pp. 1-13. (**JCR Q1; Rank: 4/72; IF: 14.919**)
36. W. Qi, W. Rosikiewicz, Z. Yin, B. Xu, H. Jiang, **S. Wan**, Y. Fan, G. Wu and L. Wang, "Genomic Profiling Identifies Genes and Pathways Dysregulated by *HEY1-NCOA2* Fusion and Shed a Light on Mesenchymal Chondrosarcoma Tumorigenesis", *Journal of Pathology*, 2022, vol. 257, no. 5, pp. 579-592. (**JCR Q1; Rank: 5/77; IF: 7.996**)
37. P. C. Chen, X. Han, T. Shaw, H. Sun, M. Niu, Z. Wang, Y. Jiao, B. Teubner, D. Eddins, L. Beloate, B. Bai, J. Mertz, Y. Li, Y. Fu, J. H. Cho, X. Wang, Z. Wu, S. Poudel, Z. F. Yuan, A. Mancieri, J. Low, H. M. Lee, M. Patton, L. Earls, E. Stewart, P. Vogel, **S. Wan**, G. Serrano, T. Beach, M. Dyer, R. Smeyne, T. Moldoveanu, T. Chen, G. Wu, S. Zakharenko, G. Yu and J. Peng, "Alzheimer's Disease-Associated U1 snRNP Splicing Dysfunction Causes Neuronal Hyperexcitability and Cognitive Impairment", *Nature Aging*, 2022, vol. 2, pp. 923-940. (**JCR Q1; Rank: 1/69; IF: 16.6**)
38. C. Jiang, **S. Wan**, P. Hu, Y. Li and S. Li, "Editorial: Transcriptional Regulation in Metabolism and Immunology", *Frontiers in Genetics*, 2022, vol. 13, article 845697. (**JCR Q2; Rank: 48/176; IF: 4.599**)
39. **S. Wan***, J. Wang, G. Wu and Y. Fan*, "Robust and Ultrafast Clustering of Large-Scale Single Cell RNA-seq Data", *Briefings in Bioinformatics*, 2022, submitted. (**JCR Q1; Rank: 3/78; IF: 11.622**)
40. S. Singh, W. Quarni, M. Goralski, **S. Wan**, H. Jin, L. A. Van de Velde, J. Fang, R. Sing, Y. Fan, M. Johnson, W. Akers, P. Murray, P. G. Thomas, D. Nijhawan, A. M. Davidoff and J. Yang, "Targeting the Spliceosome through RBM39 Degradation Results in Exceptional Responses in High-Risk Neuroblastoma Models", *Science Advances*, 2021, vol. 7, no. 47, eabj5405. (**JCR Q1; Rank: 5/73; IF: 14.136**)
41. A. Lavado, R. Gangwar, J. Pare, **S. Wan**, Y. Fan and X. Cao, "YAP/TAZ Maintain the Proliferative Capacity and Structural Organization of Radial Glial Cells during Brain Development", *Developmental Biology*, 2021, vol. 480, pp. 39-49. (**JCR Q2; Rank: 15/41; IF: 3.582**)
42. **S. Wan*** and J. Q. Wang, "KLNP-Chlo: Multi-Label Ensemble Kernelized Neighbourhood Propagation for Predicting Localization of Chloroplast Proteins", *Bioinformatics*, 2021, submitted. (**JCR Q1; Rank: 3/58; IF: 6.937**)
43. **S. Wan***, J. Z. Hu and J. Q. Wang, "Improving Prediction of Multi-Location Protein Subcellular Localization with Confidence-Weighted Gene Ontology Features", *Bioinformatics*, 2021, submitted. (**JCR Q1; Rank: 3/58; IF: 6.937**)
44. M. Zanin, S. S. Wong, P. Schreiner, **S. Wan**, P. Vogel, Z. Cheng, R. El-Shesheny, D. F. Boyd, J. C. Jones, J. DeBeauchamp, Z. A. Kocer, P. Thomas, S. Barman, T. C. Chang, Y. Fan, R. Webster, R. Webby, "Airborne Transmission of Influenza Viruses is Associated with a Distinct Host Transcriptomic Response in the Nasal Epithelium", *Proceedings of the National Academy of Sciences of the United States of America (PNAS)*, 2021, submitted. (**JCR Q1; Rank: 8/73; IF: 11.205**)
45. C. Jablonowski, G. Wu, W. Quarni, H. Hu, S. Singh, D. Bostanthirige, H. Jin, T. C. Chang, **S. Wan**, D. Finkelstein, J. H. Cho, D. Hu, V. Pagala, S. Miller, R. Wang, A. Murphy, K. Freeman, A. B. Shyu, A. M. Davidoff, J. Peng, J. Yang,

- “Metabolic Reprogramming of Cancer Cells by JMJD6-Mediated Alternative Splicing”, *Science Advances*, 2021, submitted. (*JCR Q1; Rank: 5/73; IF: 14.136*)
46. T. I. Shaw, Y. Li, H. Wang, A. Gout, X. Von Buttlar, S. Wan, X. Ma, A. C. Tan, “Analytical Challenges of Alternative Splicing in Cancer”, *Frontiers in Oncology*, 2021, submitted. (*JCR Q2; Rank: 62/242; IF: 6.244*)
47. S. Wan, J. Kim and K. J. Won, “SHARP: Hyper-Fast and Accurate Processing of Single-Cell RNA-seq Data via Ensemble Random Projection”, *Genome Research*, 2020, vol. 30, pp. 205-213. (*JCR Q1; Rank: 6/177; IF: 11.093*)
48. T. Sakamoto, T. Matsuura, S. Wan, D. Ryba, J. Kim, K. J. Won, L. Lai, C. Petucci, N. Petrenko, K. Musunuru, R. Vega, D. Kelly, “A Critical Role for Estrogen-Related Receptor Signaling in Cardiac Maturation”, *Circulation Research*, 2020, vol. 126, pp. 1685-1702. (*JCR Q1; Rank: 5/138; IF: 14.467*)
49. B. Ahn, S. Wan, N. Jaiswal, R. Vega, D. E. Ayer, P. M. Titchenell, X. Han, K. J. Won, and D. P. Kelly, “MondoA Coordinately Drives Muscle Lipid Accumulation and Insulin Resistance”, *JCI Insight*, 2019, 4(15): e129119. (*JCR Q1; Rank: 14/138; IF: 6.205*)
50. S. Wan* and M. W. Mak*, “Predicting Subcellular Localization of Multi-Location Proteins by Improving Support Vector Machines with Adaptive-Decision Schemes”, *International Journal of Machine Learning and Cybernetics*, 2018, vol. 9, pp. 399–411. (*JCR Q1; Rank: 31/134; IF: 3.844*)
51. S. Wan*, M. W. Mak*, and S. Y. Kung, "FUEL-mLoc: Feature-Unified Prediction and Explanation of Multi-Localization of Cellular Proteins in Multiple Organisms", *Bioinformatics*, 2017, vol. 33, no. 5, pp. 749–750. (*JCR Q1; Rank: 3/58; IF: 6.937*)
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53. S. Wan*, M. W. Mak*, and S. Y. Kung, "Gram-LocEN: Interpretable Prediction of Multi-Location Gram-Positive and Gram-Negative Bacterial Protein Subcellular Localization ", *Chemometrics and Intelligent Laboratory Systems*, 2017, vol. 162, pp. 1–9. (*JCR Q1; Rank: 11/123; IF: 2.895*)
54. J. Q. Wang, C. C. Zhang, S. Wan and G. Peng, "Is Congenital Amusia a Connectome Disorder?: A Diffusion MRI Study Combining Tract- and Network-Based Analysis", *Frontiers in Human Neurosciences*, 2017, vol. 11, pp. 473. doi: 10.3389/fnhum.2017.00473. (*JCR Q2; Rank: 24/77; IF: 2.673*)
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64. **S. Wan**, M. W. Mak, and S. Y. Kung, "Semantic Similarity over Gene Ontology for Multi-Label Protein Subcellular Localization", *Engineering*, 2013, vol. 5, pp. 68–72.
65. **S. Wan**, M. W. Mak, and S. Y. Kung, "GOASVM: A Subcellular Location Predictor by Incorporating Term-Frequency Gene Ontology into the General Form of Chou's Pseudo-Amino Acid Composition", *Journal of Theoretical Biology*, 2013, vol. 323, pp. 40–48. (*JCR Q1*; *Rank*: 13/57; *IF*: 2.113)
66. **S. Wan**, M. W. Mak, and S. Y. Kung, "mGOASVM: Multi-Label Protein Subcellular Localization Based on Gene Ontology and Support Vector Machines", *BMC Bioinformatics*, 2012, 13:290. **Highly accessed** (*JCR Q1*; *Rank*: 9/59; *IF*: 3.242)

CONFERENCE PAPERS

67. Z. Xu, M. Azzam, M. Bivens, M. Dustin, J. U. Blackford, **S. Wan** and J. Wang*, "nnU-BNST: Deep Learning-Based Automated Segmentation of the Bed Nucleus of the Stria Terminalis", *Machine Learning in Medical Imaging (MLMI 2025)*. Lecture Notes in Computer Science, vol. 16241, pp. 637–648.
68. **S. Wan***, J. Kim, Y. Fan and K. J. Won*, "Hyper-Fast and Accurate Clustering of Ultra-Large-Scale Single-Cell Data with Ensemble Random Projection", *The 2020 International Conference on Machine Learning (ICML) Workshop on Computational Biology*, virtual online, Jul. 2020. (**Highlights**)

Curriculum Vitae

69. **S. Wan***, J. Kim, Y. Fan and K. J. Won*, “Processing millions of single cells by SHARP”, *The 11th ACM Conference on Bioinformatics, Computational Biology and Health Informatics (ACM BCB 2020)*, virtual online, Sep. 2020. (Highlights)
70. **S. Wan**, M. W. Mak and S. Y. Kung, "Protecting Genomic Privacy by a Sequence-Based Obfuscation Method", 2017, arXiv preprint, arXiv: 1708.02629.
71. Mert Al, **S. Wan** and S. Y. Kung, "Ratio Utility and Cost Analysis for Privacy Preserving Subspace Projection", 2017, arXiv preprint, arXiv:1702.07976.
72. **S. Wan**, M. W. Mak, B. Zhang, Y. Wang and S. Y. Kung, "Ensemble Random Projection for Multi-Label Classification with Application to Protein Subcellular Localization", *2014 IEEE International Conference on Acoustic Speech and Signal Processing (ICASSP'14)*, Florence, Italy, May 2014, pp. 6040-6044.
73. **S. Wan**, M. W. Mak, B. Zhang, Y. Wang and S. Y. Kung, "An Ensemble Classifier with Random Projection for Predicting Multi-Label Protein Subcellular Localization", *The 2013 IEEE International Conference on Bioinformatics and Biomedicine (BIBM'2013)*, Shanghai, China, Dec. 2013, pp. 35-42.
74. **S. Wan**, M. W. Mak, and S. Y. Kung, "Adaptive Thresholding for Multi-Label SVM Classification with Application to Protein Subcellular Localization Prediction", *2013 IEEE International Conference on Acoustic Speech and Signal Processing (ICASSP'13)*, Vancouver, Canada, May 2013, pp. 3547-3551.
75. **S. Wan**, M. W. Mak, and S. Y. Kung, "GOASVM: Protein Subcellular Localization Prediction Based on Gene Ontology Annotation and SVM", *2012 IEEE International Conference on Acoustic Speech and Signal Processing (ICASSP'12)*, Kyoto, Japan, Mar. 2012, pp. 2229-2232.
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77. **S. Wan**, C. Yao, Y. Hu, and G. Zhang, “A Method of Continuous Data Flow Embedded within Speech Signals”, *The 2-nd International Conference on Signal Acquisition and Processing (ICSAP'10)*, Bangalore, India, Feb. 2010, pp. 362-365.

CONFERENCE ABSTRACTS

78. B. Tuerhanbayi, J. Wang, and **S. Wan***, “Integrated Meta-Clustering Reveals Subtype-Specific Transcriptomic Heterogeneity of Pediatric Low-Grade Glioma”, *Cancer Research*, 2026. (To be presented in *AACR Annual Meeting 2026*, San Diego, CA, Apr. 2026)
79. M. J. Baek, J. Wang, and **S. Wan***, “DAG-Guided Kernel Learning Framework to Mitigate Racial Disparities in Breast Cancer”, *Cancer Research*, 2026. (To be presented in *AACR Annual Meeting 2026*, San Diego, CA, Apr. 2026)
80. L. Li, J. Wang and **S. Wan***, “OTTER: Optimal Transport-Based Transcriptomics and Genomics Representation Fusion for T-ALL Subtyping”, *Cancer Research*, 2026. (To be presented in *AACR Annual Meeting 2026*, San Diego, CA, Apr. 2026)

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81. N. Peterson, J. Wang and S. Wan*, “Exploring High-Resolution Subtypes for Pancreatic Ductal Adenocarcinoma via a Meta-Clustering Approach”, *Cancer Research*, 2026. (To be presented in *AACR Annual Meeting 2026*, San Diego, CA, Apr. 2026)
82. M. Sun, J. Wang and S. Wan*, “iS3RGs: Discriminating Diverse Medulloblastoma Subtypes by Leveraging Heterogenous Transcriptome Data with Batch Effects”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 5022-5022. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)
83. L. Li, J. Khoury, J. Wang and S. Wan*, “AttentionAML: An Attention-Based Deep Learning Model for Accurate Identification of Childhood Acute Myeloid Leukemia Subtypes”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 7425-7425. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)
84. N. Peterson, M. Sun, J. Wang and S. Wan*, “Identifying Pancreatic Cancer Subtypes by a Novel Meta-Learning Model”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 1104-1104. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)
85. X. Wu, J. Wang and S. Wan*, “Combining Random Projection and Stacking Learning for NSCLC Subtype Classification Based on Transcriptomic Data”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 1098-1098. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)
86. M. J. Baek, L. Li, J. Wang, V. Band and S. Wan*, “A Weighted Multi-Modal Transfer Learning Model for Alleviating Racial Disparities in Breast Cancer”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 3620-3620. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)
87. B. Kennedy, M. Raza, S. Mirza, A. Rajan, F. Oruji, Ma. Storck, S. Lele, T. Reznicek, L. Li, J. Rowley, S. Wan, B. Mohapatra, H. Band and V. Band*, “ECD Co-Operates with ERBB2 to Promote Tumorigenesis through Upregulation of Unfolded Protein Response and Glycolysis”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 267-267. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)
88. B. Mohapatra, A. Bhat, M. Raza, H. Luan, S. Shrestha, S. Kolluru, M. Storck, L. Li, F. Kuo, S. Leit, F. Qiu, S. Lele, D. Ciccone, C. Loh, S. Wan, V. Band and H. Band*, “Targeting CBL and CBLB Ubiquitin Ligases to Exhaust Cancer Stem Cells in Metastatic Breast Cancer”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 5697-5697. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)
89. R. Liu, S. Wang, S. Wan and J. Wang*, “Enhancing Prostate Pelvic Multimodality Data Generating with Conditional Generative Models: A Pix2Pix-Based Approach for MRI-to-PET Synthesis”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 5016-5016. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)
90. M. Azzam, H. Leuva, M. Zhou, B. Teply, R. Bergan, S. Bates, S. Wan, A. Fojo and J. Wang*, “Development of G-Rate Based Random Forest Machine Learning Model to Predict Overall Survival for Patients with Metastatic Prostate Cancer”, *Cancer Research*, 2025, vol. 85 (8_Supplement_1), pp. 6314-6314. (presented in *AACR Annual Meeting 2025*, Chicago, IL, Apr. 2025)

Curriculum Vitae

91. M. Sun, J. Wang and S. Wan*, “Identifying Antimicrobial Peptides Based on Proportionalized Split Amino Acid Compositions”, *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston TX, Sep. 2024.
92. M. Sun, J. Wang and S. Wan*, “Discriminating Diverse Medulloblastoma Subtypes by Leveraging Heterogenous Transcriptome Data with Batch Effects”, *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston TX, Sep. 2024.
93. L. Li, J. Khoury, J. Wang and S. Wan*, “Pediatric Acute Myeloid Leukemia Classification via Multi-Modal Ensemble Learning”, *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston TX, Sep. 2024.
94. H. Xiao, J. Wang and S. Wan*, “WIMOAD: Weighted Integration of Multi-Omics Data for Alzheimer’s Disease (AD) Diagnosis”, *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston TX, Sep. 2024.
95. M. Sun, J. Wang and S. Wan*, “iS3RGs: Discriminating Diverse Medulloblastoma Subtypes by Leveraging Heterogenous Transcriptome Data with Batch Effects”, *Pediatric Cancer Research Symposium*, Omaha, NE, Aug. 2024.
96. L. Li, J. Khoury, J. Wang and S. Wan*, “AttentionAML: An Attention-Based Deep Learning Model for Identifying Acute Myeloid Leukemia Subtypes”, *Pediatric Cancer Research Symposium*, Omaha, NE, Aug. 2024.
97. L. Li, H. Xiao, J. Khoury, J. Wang and S. Wan*, “RanBALL: Identifying B-Cell Acute Lymphoblastic Leukemia Subtypes Based on an Ensemble Random Projection Model”, *Cancer Research*, 2024, vol. 84 (6_Supplement), pp. 4907-4907. (presented in *AACR Annual Meeting 2024*, San Diego, CA, Apr. 2024)
98. L. Li, J. Wang and S. Wan*, “Reducing Health Disparities for Prostate Adenocarcinoma by Integrating Multi-Omics Data via a Multi-Modal Transfer Learning Approach”, *Cancer Research*, 2024, vol. 84 (6_Supplement), pp. 4800-4800. (presented in *AACR Annual Meeting 2024*, San Diego, CA, Apr. 2024)
99. L. Li, H. Xiao and S. Wan*, “B-Cell Acute Lymphoblastic Leukemia Subtype Identification with an Ensemble Random Projection-Based Machine Learning Model”, *CHRI Scientific Conference*, Omaha, NE, Nov. 2023.
100. L. Li and S. Wan*, “Integrating Multi-Omics Data by a Multi-Modal Transfer Learning Model to Reduce Healthcare Disparities for Kidney Renal Clear Cell Carcinoma”, *CHRI Scientific Conference*, Omaha, NE, Nov. 2023.
101. J. Feng, M. Sun, W. Zhang, G. Wang and S. Wan*, “SAMP: An Accurate Ensemble Model Based on Proportionalized Split Amino Acid Composition for Identifying Antimicrobial Peptides”, *Antimicrobial Peptides: Yesterday, Today and Tomorrow*, Omaha, NE, Oct. 2023.
102. L. Li, H. Xiao and S. Wan*, “RanBALL: an Ensemble Random Projection-Based Model for Identifying B-Cell Acute Lymphoblastic Leukemia Subtypes”, *PCRG Symposium 2023*, Omaha, NE, Aug. 2023.
103. W. Qi, W. Rosikiewicz, Z. Yin, B. Xu, S. Wan, Y. Fan, G. Wu and L. Wang, “RNA-seq and ChIP-seq profiling identifies genes and pathways dysregulated by HEY1-NCOA2 fusion and shed a light on mesenchymal chondrosarcoma

Curriculum Vitae

- tumorigenesis”, *Cancer Research*, 2021, vol. 81 (13_Supplement), pp. LB188-LB188. (presented in *AACR Annual Meeting 2021*, Philadelphia, PA, Apr. 2021)
104. T. Sakamoto, **S. Wan**, K. Batmanov, and D. P. Kelly, "The Estrogen-Related Receptor (ERR) Drives Cardiac Myocyte Maturation in Cooperation With GATA4", *Circulation Research*, 2020, vol. 127 (Suppl_1), pp. A222-A222.
105. **S. Wan**, J. Kim, K. J. Won, and Y. Fan, “Hyper-Fast and Accurate Clustering of Ultra-Large-Scale Single-Cell Data with Ensemble Random Projection”, *Cell Symposia: The Conceptual Power of Single-Cell Biology*, San Francisco, CA, USA, Apr. 2020.
106. T. Sakamoto, **S. Wan**, K. J. Won, and D. P. Kelly, “The Estrogen-Related Receptor Coordinates Transcription of Genes Involved in Mitochondrial and Contractile Maturation in Human Induced Pluripotent Stem Cell-Derived Cardiac Myocytes”, *Circulation*, 2019, vol. 140 (Suppl_1), pp. A11803-11803. (presented in *American Heart Association Scientific Sessions (AHA2019)*, Philadelphia, PA, USA, Nov. 2019)
107. T. R. Matsuura, T. Sakamoto, D. M. Ryba, **S. Wan**, and D. P. Kelly, “Estrogen-Related Receptor Signaling Is Critical for Postnatal Cardiac Maturation”, *Circulation*, 2019, vol. 140 (Suppl_1), A13898-A13898. (presented in *American Heart Association Scientific Sessions (AHA2019)*, Philadelphia, PA, USA, Nov. 2019)
108. B. Ahn, **S. Wan**, K. J. Won, N. Jaiswal, P. M. Titchenell, and D. P. Kelly, “MondoA Mediates Myocyte Lipid Accumulation and Insulin Resistance Driven by Chronic Nutrient Excess”, *American Diabetes Association's 79th Scientific Sessions*, San Francisco, CA, USA, Jun. 2019. (oral)
109. **S. Wan**, J. Kim and K. J. Won, "Hyper-Fast and Accurate Processing of Large-Scale Single-Cell Transcriptomics Data via Ensemble Random Projection", *RECOMB/ISCB Conference on Regulatory & Systems Genomics with DREAM Challenges (RSG DREAM 2018)*, New York City, NY, USA, Dec. 2018.

BIOINFORMATICS SOFTWARE TOOLS

1. **RPSLearner**: Combining Random Projection and Stacking Learning for Lung Cancer Subtyping
<https://github.com/wan-mlab/RPSLearner>
2. **RaMBat**: A Gene-Ranking Framework to Identify Medulloblastoma Subtypes Based on Heterogeneous Transcriptomics Data with Severe Batch Effects
<https://github.com/wan-mlab/RaMBat>
3. **RanBALL**: An Ensemble Random Projection Model to Identify Subtypes of B-cell Acute Lymphoblastic Leukemia
<https://github.com/wan-mlab/RanBALL>
4. **WIMOAD**: Weighted Integration of Multi-Omics data for Alzheimer’s Disease (AD) Diagnosis
<https://github.com/wan-mlab/WIMOAD>
5. **SAMP**: A Proportionalized Split Amino Acid Composition Based Model for Identifying Antimicrobial Peptides
<https://github.com/wan-mlab/SAMP>

Curriculum Vitae

6. **AttentionAML**: An Attention-based MLP Model for Identifying Acute Myeloid Leukemia Subtypes
<https://github.com/wan-mlab/AttentionAML>
7. **SHARP**: Single-cell RNA-seq Hyper-fast and Accurate processing via ensemble Random Projection
<https://github.com/shibiaowan/SHARP>
8. **IterMegaBLAST**: A Sequence-Similarity Based Obfuscation Method for Genomic Privacy Protection
<https://github.com/shibiaowan/IterMegaBLAST>
9. **FUEL-mLoc**: Feature-Unified Prediction and Explanation of Protein Multi-Localization of Cellular Proteins (for Eukaryote, Human, Plant, Gram-positive, Gram-negative and Virus)
<http://bioinfo.eie.polyu.edu.hk/FUEL-mLoc/>
10. **Gram-LocEN**: Interpretable Prediction of Subcellular Multi-Localization of Gram-Positive and Gram-Negative Bacterial Proteins (for Gram-positive Bacteria, Gram-negative Bacteria)
<http://bioinfo.eie.polyu.edu.hk/Gram-LocEN/>
11. **PolyU-Loc**: A Package of Web-Servers for Protein Subcellular Localization Prediction
http://bioinfo.eie.polyu.edu.hk/Book_website/
12. **EnTrans-Chlo**: Transductive Learning for Protein Subchloroplast Localization Prediction
<http://bioinfo.eie.polyu.edu.hk/EnTransChloServer/>
13. **LNP-Chlo**: Linear Neighborhood Propagation for Protein Subchloroplast Localization Prediction
<http://bioinfo.eie.polyu.edu.hk/LNPChloServer/>
14. **Mem-ADSVM**: A Two-Layer Predictor for Multi-Label Membrane Protein Type Prediction
<http://bioinfo.eie.polyu.edu.hk/MemADSVMServer/>
15. **Mem-mEN**: A Predictor for Membrane Protein Multi-Functional Type Prediction
<http://bioinfo.eie.polyu.edu.hk/MemmENServer/>
16. **SpaPredictor**: An Interface of Two Predictors for Interpretable Protein Subcellular Localization
<http://bioinfo.eie.polyu.edu.hk/SpaPredictorServer/>
17. **HybridGO-Loc**: Mining Hybrid GO Features for Protein Subcellular Localization Prediction (for Virus and Plant)
<http://bioinfo.eie.polyu.edu.hk/HybridGoServer/>
18. **R3P-Loc**: Compact Predictor for Multi-Label Protein Subcellular Localization (for Eukaryote and Plant)
<http://bioinfo.eie.polyu.edu.hk/R3PLocServer/>
19. **mPLR-Loc**: Probabilistic Predictor for Multi-label Protein Subcellular Localization (for Virus and Plant)
<http://bioinfo.eie.polyu.edu.hk/mPLRLocServer/>
20. **mLASSO-Hum**: An Interpretable Predictor for Human Protein Subcellular Localization
<http://bioinfo.eie.polyu.edu.hk/mLASSOHumServer/>
21. **mGOASVM**: Multi-label Protein Subcellular Localization Prediction (for Virus and Plant)
<http://bioinfo.eie.polyu.edu.hk/mGoaSvmServer/mGOASVM.html>

Curriculum Vitae

22. GOASVM: Single-label Protein Subcellular Localization Prediction (for Eukaryote and Human)

<http://bioinfo.eie.polyu.edu.hk/mGoaSvmServer/GOASVM.html>

INVITED TALKS

1. “AI and Bioinformatics for Cancer Research”, in *FPBCC Cancer Biology Program Monthly Meeting of University of Nebraska Medical Center*, Omaha, NE, USA, October 2025.
2. “AI for Pediatric Cancer Research”, invited as a **Featured Speaker** for the *2025 Pediatric Cancer Research Group Symposium (PCRG)*, Omaha, NE, USA, August 2025.
3. “Leveraging AI for Characterizing Pediatric Cancer”, invited as a **Featured Speaker** for *The 2025 International Conference on Intelligent Biology and Medicine (ICIBM 2025)*, Columbus, OH, USA, August 2025.
4. “Advancing Biomedical Research by AI and Bioinformatics”, in *State Key Laboratory of Primate Biomedical Research of Kunming University of Science and Technology*, Kunming, China, June 2025.
5. “Leveraging AI for Characterizing Pediatric Cancer”, in *CHRI Artificial Intelligence (AI) Virtual Seminar Series of Child Health Research Institute (CHRI)*, Omaha, NE, USA, May 2025.
6. “AI and Bioinformatics for Omics-Based Biomedical Research”, in *Wyoming Sensory Biology Center of Biomedical Research Excellence (WSBC) and Department of Zoology and Physiology of University of Wyoming*, Laramie, WY, USA, April 2025.
7. “Advancing Biomedical Research by Artificial Intelligence and Bioinformatics”, invited as a **Featured Speaker** for the *1st Annual Basic Biomedical Research Retreat of University of Nebraska Medical Center*, Omaha, NE, USA, March 2025.
8. “Machine Learning for Large-Scale Single Cell Analysis”, in *Immunology Interest Group of University of Nebraska Medical Center*, Omaha, NE, USA, February 2025.
9. “Artificial Intelligence for Omics-Based Biomedical Research”, invited as a **Featured Speaker** for *The 2024 International Conference on Intelligent Biology and Medicine (ICIBM 2024)*, Houston, TX, USA, October 2024.
10. “Analyzing Millions of Single Cells via SHARP”, invited as a **Featured Speaker** for *Bioinformatics Research Symposium and Workshop on Integrative Omics and ML*, Lincoln, NE, USA, August 2024.
11. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Center for Aquacultural Breeding Research of Shanghai Ocean University*, Shanghai, China, July 2024.
12. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Department of Electrical and Electronic Engineering of The Hong Kong Polytechnic University*, Hong Kong SAR, China, July 2024.
13. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Health Data Science Center of Shenzhen People’s Hospital*, Shenzhen, China, July 2024.
14. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Institute of Artificial Intelligence of Wuhan University*, Wuhan, China, June 2024.

Curriculum Vitae

15. “Artificial Intelligence for Omics-Based Biomedical Research”, in *School of Artificial Intelligence and Automation of Huazhong University of Science and Technology*, Wuhan, China, June 2024.
16. “RanBALL: A Machine Learning Model Based on Ensemble Random Projection for Accurate Identification of B-Cell Acute Lymphoblastic Leukemia Subtypes”, in *Hematological Malignancies Group of University of Nebraska Medical Center*, Omaha, NE, USA, June 2024.
17. “Artificial Intelligence for Omics-Based Biomedical Research”, invited as a **Featured Speaker** for the Symposium on *ML/AI Applications in Biology and Medicine*, Omaha, NE, USA, May 2024.
18. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Department of Animal Sciences of University of Nebraska-Lincoln*, Lincoln, NE, USA, March 2024.
19. “Machine Learning for Omics-Based Biomedical Research”, in *VA Research Seminar of VA Nebraska-Western Iowa Health Care System*, Omaha, NE, USA, January 2024.
20. “Artificial Intelligence for Omics-Based Biomedical Research”, in *Surgery Research Forum of Department of Regenerative Medicine of University of Nebraska Medical Center*, Omaha, NE, USA, January 2024.
21. “Machine Learning for Processing Large-Scale Biological Data”, invited as a **Keynote Speaker** for *The 2023 Midlands Society of Physiological Sciences (MSPS) Annual Meeting*, Omaha, NE, USA, October 2023.
22. “Machine Learning for Large-Scale Bioinformatics Analysis”, in *Department of Biochemistry and Molecular Biology of University of Nebraska Medical Center*, Omaha, NE, USA, June 2023.
23. “Machine Learning for Processing Large-Scale Biological Data”, in *Department of Biostatistics of University of Nebraska Medical Center*, Omaha, NE, USA, January 2023.
24. “Machine Learning for Processing Large-Scale Biological Data”, in *Department of Genetics, Cell Biology and Anatomy of University of Nebraska Medical Center*, Omaha, NE, USA, May 2022.
25. “Machine Learning for Processing Large-Scale Biological Data”, in *Department of Chemistry and Chemical Biology of Stevens Institute of Technology*, Hoboken, NJ, USA, April 2022.
26. “Machine Learning and Data Science for Biomedicine and Cancer Research”, in *BNU-HKBU United International College*, Zhuhai, China, April 2022.
27. “Machine Learning for Large-Scale Single-Cell Data Processing”, in *Department of Chemical Biology and Therapeutics of St. Jude Children’s Research Hospital*, Memphis, TN, USA, February 2021.
28. “Supervised Learning”, in *Center for Applied Bioinformatics of St. Jude Children’s Research Hospital*, Memphis, TN, USA, January 2021.
29. “Introduction to Machine Learning”, in *Center for Applied Bioinformatics of St. Jude Children’s Research Hospital*, Memphis, TN, USA, November 2020.
30. “SHARP: Single-Cell RNA-seq Hyper-Fast and Accurate Clustering via Ensemble Random Projection”, in *School of Computer Science and Technology of Xidian University*, Xi’an, China, August 2020.
31. “Assessing Gene Expression Dynamics with SLAM-Seq”, in *Center for Applied Bioinformatics of St. Jude Children’s Research Hospital*, Memphis, TN, USA, May 2020.

Curriculum Vitae

32. “SHARP: Hyper-fast and Accurate Processing of Single-Cell RNA-Seq Data via Ensemble Random Projection”, in *Center for Applied Bioinformatics of St. Jude Children’s Research Hospital*, Memphis, TN, USA, February 2020.
33. “SHARP: Single-Cell RNA-Seq Hyper-fast and Accurate Processing via Ensemble Random Projection”, in *Single-Cell Working Group at Smilow Center for Translational Research of University of Pennsylvania*, Philadelphia, PA, USA, February 2019.
34. “Machine Learning for Bioinformatics Data Science”, in *Department of Computational Biology of St. Jude Children’s Research Hospital*, Memphis, TN, USA, January 2019.
35. “Machine Learning for Bioinformatics Data Science: A Study on Protein Subcellular Localization”, in *Department of Electronic Engineering of City University of Hong Kong*, Hong Kong SAR, China, June 2017.
36. “Machine Learning for Predicting Protein Subchloroplast Localization”, in *Department of Computer Science of University of Bristol*, Bristol, UK, June 2017.
37. “Data Science in Bioinformatics: A Study on Protein Subcellular Localization”, in *Center for Big Data Research of Tsinghua-Berkeley Shenzhen Institute*, Shenzhen, China, May 2017.
38. “Predicting Protein Subcellular Localization by Bioinformatics Approaches”, in *Department of Pediatrics of The Children’s Hospital of Philadelphia*, Philadelphia, PA, USA, April 2017.
39. “Combining Feature-Selection and Subspace Approaches for Utility-Privacy Tradeoff”, in *Department of Electrical Engineering of Princeton University*, Princeton, NJ, USA, March 2017.
40. “Bioinformatics Approaches for Protein Subcellular Localization Prediction”, in *Institute for Diabetes, Obesity and Metabolism of Perelman School of Medicine at University of Pennsylvania*, Philadelphia, PA, USA, March 2017.
41. “Linear and Kernelized Neighborhood Propagation: A Semi-Supervised Approach”, in *Department of Electrical Engineering of Princeton University*, Princeton, NJ, USA, December 2016.
42. “Semi-Supervised Multi-Kernel Learning in Image Classification”, in *Department of Electrical Engineering of Princeton University*, Princeton, NJ, USA, October 2016.
43. “Large-Scale in-Silico Prediction of Protein Subcellular Localization”, in *the 3-rd Wuhan University International Forum for Interdisciplinary Sciences and Engineering*, in *School of Electronic Information of Wuhan University*, Wuhan, China, April 2016.
44. “Machine Learning for Multi-Label Protein Subcellular Localization”, in *Department of Computer Science of Hong Kong Baptist University*, Hong Kong SAR, China, May 2015.
45. “Machine Learning Approaches for Protein Subcellular Localization”, in *Department of Statistics of The Chinese University of Hong Kong*, Hong Kong SAR, China, Apr. 2015.
46. “CPTAC Sub-Project: Customized Sample-Specific Protein Sequence Database”, in *Computational Bioinformatics and Bioimaging Laboratory (CBIL) of Virginia Tech*, VA, USA, May 2013.
47. “mGOASVM: A Multi-Label Protein Subcellular Localization Predictor”, in *Computational Bioinformatics and Bioimaging Laboratory (CBIL) of Virginia Tech*, VA, USA, May 2013.

Curriculum Vitae

48. “Semantic Similarity over Gene Ontology for Multi-label Protein Subcellular Localization”, in *2013 International Conference on Bioinformatics and Biomedical Engineering (iCBBE2013)*, Beijing, China, Sep. 2013.
49. “Protein Subcellular Localization Prediction Based on Profile Alignment and Gene Ontology”, in *the 6-th Beijing-Hong Kong International Doctoral Forum 2011*, Hong Kong SAR, China, Aug. 2011.
50. “Support Vector Data Description (SVDD) vs Support Vector Machine (SVM)”, in *Center for Multimedia Signal Processing of The Hong Kong Polytechnic University*, Hong Kong SAR, China, Mar. 2011.

AWARDS & HONORS

- **Springer Nature Editor of Distinction Award** for *BMC Bioinformatics*, Springer Nature, 2025;
- **Certificate of Achievement in Research**, awarded by UNMC College of Medicine, Omaha, NE, USA, 2025;
- **New Investigator Award**, awarded by UNMC, Omaha, NE, USA, 2024;
- **FIRST Award of Nebraska EPSCoR Program**, Lincoln, NE, USA, 2023;
- **IEEE Senior Member**, Piscataway, NJ, USA, 2022;
- **EIE Outstanding Young Alumni Award**, Hong Kong SAR, China, 2022;
- **Top reviewers in Cross-Field (top 1%)**, awarded by Clarivate, USA, 2019;
- **Top reviewers in Biology and Biochemistry (top 1%)**, awarded by Clarivate, USA, 2019;
- **Brandeis Postdoctoral Fellowship**, Princeton, NJ, USA, 2016-2018;
- **“Best Poster Award”** in *the 6-th Beijing-Hong Kong International Doctoral Forum 2011*, Hong Kong SAR, August, 2011;
- **Postgraduate Scholarship**, The Hong Kong Polytechnic University, Hong Kong SAR, 2010-2014;
- **National Scholarship**, Wuhan, China, 2008-2009;
- **National Inspirational Scholarship**, Wuhan, China, 2007-2008;
- **Excellent Scholarship** (the 3-rd prize), Wuhan, China, 2007-2008;
- **The Freshmen Scholarship**, Wuhan, China, 2006-2007.

TEACHING COURSES

- **2025**
 - *Tools and Algorithms in Bioinformatics* (BISB815), University of Nebraska Medical Center
 - *Bioinformatics in Scientific Publications* (BISB903), University of Nebraska Medical Center
 - *Modern Approaches in Cell Biology and Molecular Genetics* (MGCB912), University of Nebraska Medical Center
 - *Principles and Methodologies of Cancer Research* (CRGP880), University of Nebraska Medical Center
- **2024**
 - *Tools and Algorithms in Bioinformatics* (BISB815), University of Nebraska Medical Center

Curriculum Vitae

- *Modern Approaches in Cell Biology and Molecular Genetics* (MGCB912), University of Nebraska Medical Center
- *Principles and Methodologies of Cancer Research* (CRGP880), University of Nebraska Medical Center
- **2023**
 - *Tools and Algorithms in Bioinformatics* (BISB815), University of Nebraska Medical Center
 - *Bioinformatics in Scientific Publications* (BISB903), University of Nebraska Medical Center
 - *Modern Approaches in Cell Biology and Molecular Genetics* (MGCB912), University of Nebraska Medical Center
 - *MGCB & BISB Fellowship Writing*, University of Nebraska Medical Center
- **2019-2022**
 - *Machine Learning for Applied Bioinformatics*, St Jude Children's Research Hospital
 - *Journal Club for Machine Learning for Applied Bioinformatics*, St Jude Children's Research Hospital
- **2017**
 - *Deep Neural Computing* (ELE571), Princeton University
- **2016**
 - *Kernel-Based Machine Learning* (ELE477), Princeton University
- **2015**
 - *Object-Oriented Design and Programming* (EIE320/EIE3375), The Hong Kong Polytechnic University
- **2014**
 - *Information Technology* (ENG2003), The Hong Kong Polytechnic University
 - *Database Systems* (EIE3114), The Hong Kong Polytechnic University
- **2013**
 - *Information Technology* (ENG2003), The Hong Kong Polytechnic University
 - *Object-Oriented Design and Programming* (EIE320/EIE3375), The Hong Kong Polytechnic University
 - *Database Systems* (EIE3114), The Hong Kong Polytechnic University
 - *Distributed Systems and Network Programming* (EIE424/EIE4108), The Hong Kong Polytechnic University
- **2012**
 - *Object-Oriented Design and Programming* (EIE320/EIE3375), The Hong Kong Polytechnic University
 - *Distributed Systems and Network Programming* (EIE424/EIE4108), The Hong Kong Polytechnic University
 - *Communication Fundamentals* (EIE331), The Hong Kong Polytechnic University

MENTORED TRAINEES/POSTDOCS/STUDENTS

Curriculum Vitae

- **Trainees' Achievements from Wan Lab**

Date	Trainee's Name	Trainee's Role	Achievements
10-12-2025	Eswar Ramamurthy	Summer Undergraduate Research Program (SURP) trainee	Awarded the Best Poster Presentation from the Midlands Society of Physiological Sciences Annual Meeting 2025 .
06-30-2025	Bulidierxin Tuerhanbayi	PhD student	Awarded the Ray H. Frohn Incoming Graduate Student Scholarship
06-26-2025	Mengtao Sun	PhD student	Awarded the UNMC Graduate Fellowship 2025
05-09-2025	Hanyu Xiao	Master student	Officially graduated from UNMC and received the PhD offer for UT Austin
04-21-2025	Eswar Ramamurthy	Summer Undergraduate Research Program (SURP) trainee	Awarded the First Place Award for CHRI Research Forum Awards for Trainees and Associates
03-28-2024	Hanyu Xiao	Master student	Awarded 2024 Nancy and Ronald Reagan Alzheimer's Scholarship Fund Award

- **University of Nebraska Medical Center**

- **Graduate Students:**

- Bulidierxin Tuerhanbayi, PhD student, 08/2025 – present.
- Nicholas Peterson, master student, 08/2024 – present.
- Min-jeong Baek, PhD student, 08/2024 – present.
- Kah Meng Soh, PhD student (co-supervised with Dr. Jieqiong Wang), 08/2024 – present.
- Xinchao Wu, PhD student, 08/2024 – present.
- Hanyu Xiao, master student (Graduated in 05/2025), 08/2023 – 05/2025.
- Lusheng Li, PhD student, 08/2023 – present.
- Mengtao Sun, PhD student, 08/2023 – present.

- **Research Interns and Visiting Scholars:**

- Junxi Feng, master student in Harvard University, USA
- Andy Cong Liu, Sophomore in University of Waterloo, Canada
- Cece Zhang, Freshman in University of Toronto, Canada
- Yijin Zou, Senior in China Agricultural University, China
- Wenhao Zhang, master student in Xidian University, China
- Xuehuan Zhu, master graduate in UCLA, USA
- Simarpreet Kaur, BS-MS student in Indian Institute of Science Education and Research, India

Curriculum Vitae

- Eswar Byrav Ramathurthy, Freshman in University of Nebraska-Lincoln, USA
- Grace (Jiaqi) Ge, Freshman in Guangdong Technion-Israel Institute of Technology, China
- Weiqi Ye, Junior student in University of California at Davis, USA
- Saiyi Cao, Freshman in Guangdong Technion-Israel Institute of Technology, China
- Neil Linehan, master student in Western Governors University, USA
- Ouyang Han, master student in Northwestern University, USA
- Anulika Pearl Nwashili, master student in University of Arkansas at Little Rock, USA
- Sili Li, Junior in University of Wisconsin – Madison, USA
- Navya Pavani Pachava, master student in University of Nebraska at Omaha, USA
- Xingmin Hu, Junior student in Central University of Finance and Economics, China
- Nishitha Reddy Thummala, Master Student in University of Nebraska at Omaha, USA
- **Dr. Xiaoting Fan**, Associate Chief Physician at Linyi People's Hospital, Shandong, China
- Xingzhi Niu, Master Graduate Student in University of Washington, USA
- **Princeton University**
 - **Graduate Advisee:**
 - Mert Al (Supervisor: Prof. S. Y. Kung)
 - **Undergraduate Advisee:**
 - Artur Filipowicz (Supervisor: Prof. S. Y. Kung)

INTERNAL SERVICE WITHIN UNMC

- **Administration Committee**
 - Co-Director, BISB PhD Program, 2025 – present
 - Facilitator, University of Nebraska Systems Faculty Collaboration Initiative Annual Meeting, 2025
 - Assistant Director, Bioinformatics and Systems Biology Core, 2022 – 2025
 - Member, BISB Graduate Committee, UNMC, 2022 – 2025
 - Member, Data Science Faculty Search Committee, College of Allied Health Profession of UNMC, 2025
 - Member, AI Task Force, UNMC/Nebraska Medicine, 2023 – present
- **Graduate Student Supervisory Committee**
 - Amin Foroughi-Nezhad, Biomedical Informatics (BMI) PhD Program, UNMC
 - Fnu Aditi, Molecular Genetics and Cell Biology (MGCB) PhD Program, UNMC
 - Richard Dang, Bioinformatics and Systems Biology (BISB) PhD Program, UNMC
 - Wenyang Wang, Molecular Genetics and Cell Biology (MGCB) PhD Program, UNMC
 - Kayode Raheem, Bioinformatics and Systems Biology (BISB) PhD Program, UNMC
 - Ravneet Singh Chawla, Cancer Research (CR) PhD Program, UNMC
 - Esther Ugwueke, Bioinformatics and Systems Biology (BISB) PhD Program, UNMC

Curriculum Vitae

- Michelle Aguilar-Gaspar, Molecular Genetics and Cell Biology (MGCB) PhD Program, UNMC
- Ruobin Liu, Biomedical Informatics (BMI) PhD Program, UNMC
- Saadyeh Rashidi, Bioinformatics and Systems Biology (BISB) PhD Program, UNMC
- Eric Zadok Mpingirika, Cancer Research (CR) PhD Program, UNMC
- Ritesh Dey, Molecular Genetics and Cell Biology (MGCB) PhD Program, UNMC
- Farshid Oruji, Molecular Genetics and Cell Biology (MGCB) PhD Program, UNMC
- Ben Nolan, Bioinformatics and Systems Biology (BISB) PhD Program, UNMC
- Ashish Das, Immunology, Pathology and Infectious Diseases (IPID) PhD Program, UNMC

EXTERNAL SERVICE AND PROFESSIONAL ACTIVITIES

- **Grant Reviewer for**
 - Natural Sciences and Engineering Research Council of Canada (NSERC), Canada, 2026
 - NIH Special Emphasis Panel (SEP) in Biodata Management and Computational Modeling, 2026
 - NIH Special Emphasis Panel (SEP) in the Health Services and Systems Review Branch (HSS) on Clinical Informatics, Data Management, Digital Health, and AI/ML, 2025
 - NIH Special Emphasis Panel (SEP) in Small Business: Computational, Modeling, and Biodata Management (MCST (14)), 2025
 - Czech Science Foundation, Czech Republic, 2025
 - NIH Special Emphasis Panel (SEP) in the Health Services and Systems Review Branch (HSS) on Clinical Data Management, Clinical Informatics, and Digital Health, 2025
 - NSF Graduate Research Fellowship Program (GRFP), 2024
 - Quebec Consortium for Drug Discovery (CQDM), Canada, 2024
 - UNMC College of Public Health Innovation Fund, 2024
 - Poland National Science Center, Poland, 2024
 - Medical Research Council (MRC) of UK Research and Innovation (UKRI), UK, 2024
 - NIH RECOVER COVID Initiative Systems Biology Multiomics Study Section, 2024
 - UNMC Graduate Studies Fellowship, 2024
 - NIH Biodata Management and Analysis (BDMA) Study Section, 2023
 - NSF Graduate Research Fellowship Program (GRFP), 2023
 - NIH Early Career Reviewer (ECR) Program, 2023
 - UNMC Buffet Cancer Center Pilot Grant Review Panel, 2023
 - UNMC Graduate Studies Fellowship Study Section on Neuroscience, 2023
 - UNMC Graduate Studies Fellowship Study Section on Community & Population Health/Nursing/Clinical & Translational Research, 2023
 - UNMC Buffet Cancer Center Pilot Grant Review Panel, 2022

Curriculum Vitae

- **Editor-in-Chief** for *Current Proteomics* (SCI/JCR indexed)
- **Associate Editor/Academic Editor/Editorial Board Member** for
 - *Heliyon* (**Impact Factor: 4**; Cell Press/Elsevier)
 - *Briefings in Functional Genomics* (**Impact Factor: 4.840**; Oxford University Press)
 - *BMC Bioinformatics* (**Impact Factor: 3.328**; Springer Nature)
 - *Discover Oncology* (**Impact Factor: 2.8**; Springer Nature)
 - *PeerJ Computer Science* (**Impact Factor: 3.8**; PeerJ)
 - *International Journal of Microbiology* (**Impact Factor: 3.4**; Wiley)
 - *BioMed Research International* (**Impact Factor: 3.411**; Wiley)
 - *Computational and Mathematical Methods in Medicine* (**Impact Factor: 2.809**; Wiley)
- **Guest Associate Editor** for
 - *International Journal of Molecular Sciences* (**Impact Factor: 5.6**) Research Topic on Ferroptosis: Emerging Mechanisms, Physiological Functions, and Therapeutic Applications
 - *Discover Life* (**Springer New Journal**)
 - *Frontiers in Psychology* (**Impact Factor: 3.8**) Research Topic on Subjective Well-Being and Human Decision Behaviors
 - *Frontiers in Immunology* (**Impact Factor: 8.786**) Research Topic on New Insight of Immunosenescence and Inflammaging in Geriatric Medicine and Multimorbidity
 - *Briefings in Functional Genomics* (**Impact Factor: 4.0**) Research Topic on Artificial Intelligence and Multi-Omics Research for Human Diseases
 - *BMC Methods* (**Springer New Journal**) Research Topic on Artificial Intelligence for Omics Data Analysis
 - *BMC Bioinformatics* (**Impact Factor: 3.328**) Research Topic on Evolutionary Studies
 - *Frontiers in Genetics* (**Impact Factor: 4.599**) Research Topic on Identification of Susceptibility Genes to Long COVID
 - *Frontiers in Pharmacology* (**Impact Factor: 5.988**) Research Topic on Ferroptosis as a Novel Therapeutic Target for Inflammation-Related Diseases
 - *Frontiers in Oncology* (**Impact Factor: 5.738**)
 - *Biology* (**Impact Factor: 5.079**) Research Topic on Bioinformatics and Machine Learning for Cancer Biology (Volume II)
 - *Genes* (**Impact Factor: 4.141**) Research Topic on Machine Learning Applications in Genetics
 - *Frontiers in Genetics* (**Impact Factor: 4.599**) Research Topic on Bioinformatics Analysis of Omics Data for Biomarker Identification in Clinical Research–Volume II
 - *Frontiers in Cell and Developmental Biology* (**Impact Factor: 6.684**) Research Topic on Single Cell Meets Metabolism and Cancer Biology
 - *Biology* (**Impact Factor: 5.079**) Research Topic on Bioinformatics and Machine Learning for Cancer Biology

Curriculum Vitae

- *Frontiers in Genetics* (**Impact Factor: 4.599**) Research Topic on Transcriptional Regulation in Metabolism and Immunology
- *Frontiers in Genetics* (**Impact Factor: 4.599**) Research Topic on Bioinformatics Analysis of Omics Data for Biomarker Identification in Clinical Research
- Editorial Board Member for *The Open Bioinformatics Journal*
- Editorial Board Member for *Gene & Translational Bioinformatics (GTB)*
- Topical Advisory Panel Member for *International Journal of Molecular Sciences* (**Impact Factor: 5.6**)
- Review Editor for *Frontiers in Immunology* (**Impact Factor: 8.786**)
- Review Editor for *Frontiers in Bioinformatics*
- **International Conference Committee**
 - **Session Chair**, for *The 2026 International Conference on Intelligent Biology and Medicine (ICIBM 2026)* in Buffalo, NY, 2026.
 - **Co-Chair, Publicity Committee** for *The 2025 International Conference on Intelligent Biology and Medicine (ICIBM 2025)* in Columbus, OH, 2025.
 - **Member, Scientific Program Committee (SPC)** for *The 2025 American Medical Informatics Association (AMIA) Annual Symposium* in Atlanta, GA, 2025.
 - **Co-Organizer, Organization Committee** for *The Symposium on 2024 ML/AI Applications in Biology and Medicine* in Omaha, NE, 2024.
- **Technical Program Committee (TPC) Member for International Conferences**
 - *The 2025 International Conference on Data Analytics for Business and Industry (DATA'25)*;
 - *The 2025 International Conference on Decision Aid Sciences and Applications (DASA'2025)*;
 - *The 37th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2025)*;
 - *The 30th Asia Pacific Conference on Communications (APCC 2025)*;
 - *The 11th International Conference on Mobile Applications (MobiApps 2025)*;
 - *The 2025 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2025)*;
 - *The 2025 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2025)*;
 - *The 2025 IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS 2025)*;
 - *The 36th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2024)*;
 - *The 2024 IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS 2024)*;
 - *The 5th International Conference on Smart Sensors and Application (ICSSA 2024)*;
 - *The 10th International Conference on Mobile Applications (MobiApps 2024)*;
 - *The 2024 IEEE Asia Pacific Conference on Wireless and Mobile (APWiMob 2024)*;
 - *The 29th Asia-Pacific Conference on Communications (APCC'2024)*;

Curriculum Vitae

- *The 2024 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2024);*
- *The 2024 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2024);*
- *The 35th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2023);*
- *The 9th International Workshop on Mobile Applications (MobiApps 2023);*
- *The 2023 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2023);*
- *The 2023 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2023);*
- *The 34th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2022);*
- *The 2022 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2022);*
- *The 2022 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2022);*
- *The 2022 IEEE International Conference on Internet of Things and Intelligence Systems (IoTaIS 2022);*
- *The 2022 International Conference on Innovation and Intelligence for Informatics, Computing, and Technologies (3ICT'2022);*
- *The 2022 International Conference on Decision Aid Sciences and Applications (DASA '2022);*
- *The 33rd IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2021);*
- *The 2021 IEEE International Conference on Internet of Things and Intelligent Systems (IoTaIS 2021);*
- *The 2021 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2021);*
- *The 2021 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2021);*
- *The 2021 International Conference on Data Analytics for Business and Industry (DATA '21);*
- *2021 International Conference on Innovation and Intelligence for Informatics, Computing and Technologies (3ICT 2021);*
- *The 32nd IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2020);*
- *The Second International Conference on Big Data and Advanced Wireless Technologies (BDAW 2020)*
- *The 1st Conference on Internet of Things and Embedded Intelligence (CITEI 2020)*
- *International Journal of Computing and Digital Systems'20 (IJCDS 2020)*
- *Bulletin of Electrical Engineering and Informatics (BEEI 2020);*
- *The 7th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2020);*
- *The 2nd Ahmad Dahlan International Conference Series on Engineering, Science and Information Technology (ADICS-ESIT 2020);*
- *The International Conference on Smart Education and Applied Social Sciences (SEAS 2020);*
- *The 31st IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2019);*

Curriculum Vitae

- *The 6th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2019);*
- *International Conference on Industry 4.0 and Artificial Intelligence Technologies (INAIT) 2019;*
- *International Conference on Digital Image and Signal Processing (DISP'2019) 2019;*
- *The 1st International Conference on Computer Science, Information Technology and Electrical Engineering (ICOMITEE'2019) 2019;*
- *Society of Engineering Science (SES'19) 2019;*
- *The 30th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2018);*
- *The First IEEE International Conference on Artificial Intelligence for Industries (AI4I 2018);*
- *2018 International Conference on Computer Intelligent Systems & Networking (ICCISN 2018);*
- *The 5th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2018);*
- *21st Saudi Computer Society National Computer Conference (SCS-NCC 2018);*
- *The 1st International Conference and Workshop on Telecommunication, Computing, Electronics and Control 2018 (ICW-TELKOMNIKA 2018);*
- *The 29th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2017);*
- *The 28th IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2016);*
- **Reviewer for Journals**
 - ***Nature Biotechnology***
 - ***Nature Methods***
 - ***Nature Communications***
 - ***Nature Computational Science***
 - ***Science Advances***
 - ***Advanced Science***
 - ***ACM Computing Surveys (IF: 28; ranking: 1/147)***
 - ***Cancer Research***
 - ***Genome Biology***
 - ***Genome Medicine***
 - ***Genome Research***
 - ***Nucleic Acids Research***
 - ***Journal of the American Chemical Society***
 - ***IEEE Transactions on Neural Networks and Learning Systems (IEEE TNNLS)***
 - ***IEEE Transactions on Evolutionary Computation***
 - ***British Journal of Cancer***
 - ***Clinical and Translational Medicine***

Curriculum Vitae

- *Journal of Nanobiotechnology*
- *Aging and Diseases*
- *Bioinformatics*
- *Briefings in Bioinformatics*
- *Briefings in Functional Genomics*
- *Journal of Proteome Research*
- *PLoS Computational Biology*
- *PLoS Genetics*
- *PLoS One*
- *Cancer Medicine*
- *Knowledge-Based Systems*
- *Expert Systems with Applications*
- *iMeta*
- *Seminars in Cancer Biology*
- *Frontiers in Cell and Developmental Biology*
- *Frontiers in Immunology*
- *Frontiers in Oncology*
- *Frontiers in Microbiology*
- *Frontiers in Public Health*
- *Frontiers in Genetics*
- *Frontiers in Human Neuroscience*
- *BMC Bioinformatics*
- *Computational and Structural Biotechnology Journal*
- *Journal of Chemical Information and Modeling*
- *Cells*
- *Genomics*
- *Genes*
- *Proteomes*
- *Biochimie*
- *Biosensors*
- *Biotechnology Journal*
- *IEEE/ACM Transactions on Computational Biology and Bioinformatics (IEEE/ACM TCBB)*
- *IEEE Transactions on Nanobioscience (IEEE T-NB)*
- *IEEE Access*
- *Future Generation Computer Systems*

Curriculum Vitae

- *Journal of Theoretical Biology (JTB)*
- *Journal of Integrative Bioinformatics (JIB)*
- *Analytical Biochemistry (AB)*
- *Biology*
- *Micromachines*
- *Microorganisms*
- *Molecules*
- *Biomolecules*
- *Diagnostics*
- *Sensors*
- *BioMed Research International*
- *BMC Medical Informatics and Decision Making*
- *International Journal of Molecular Sciences (IJMS)*
- *International Journal of Machine Learning and Cybernetics (JMLC)*
- *Journal of Ambient Intelligence and Humanized Computing (AIHC)*
- *Symmetry*
- *Computational Biology and Chemistry*
- *Applied Mathematics and Computation (AMC)*
- *Journal of Applied Mathematics (JAM)*
- *Medicines*
- *Advances in Artificial Neural Systems (AANS)*
- *International Journal of Biomedical Imaging (IJBI)*
- *Computer Methods and Programs in Biomedicine (CMPB)*
- *International Journal of Disaster Risk Reduction (IJDRR)*
- *International Journal of Environmental Research and Public Health (IJERPH)*
- *Applied Sciences*
- *Electronics*
- *Cancers*
- **Reviewer for International Conferences**
 - *2025 International Joint Conference on Neural Networks (IJCNN 2025);*
 - *2024 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2024);*
 - *2023 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2023);*
 - *2022 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2022); (8 times)*
 - *2021 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2021); (8 times)*

Curriculum Vitae

- *The 2021 IEEE International Conference on Industry 4.0, Artificial Intelligence, and Communication Technology (IAICT 2021)*; (2 times)
- *The 2021 IEEE Symposium on Future Telecommunication Technologies (SOFTT'2021)*; (1 time)
- *2020 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2020)*; (7 times)
- *The 1st Conference on Internet of Things and Embedded Intelligence (CITEI 2020)*; (2 times)
- *Bulletin of Electrical Engineering and Informatics (BEEI 2020)*; (2 times)
- *2019 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2019)*; (8 times)
- *The 1st International Conference on Computer Science, Information Technology and Electrical Engineering (ICOMITEE'2019)*; (1 time)
- *The 6th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2019)*; (1 time)
- *2018 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2018)*; (3 times)
- *The First IEEE International Conference on Artificial Intelligence for Industries (AI4I 2018)*; (2 times)
- *The 5th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI 2018)*; (2 times)
- *2017 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP2017)*; (3 times)
- *2017 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2017)*; (1 time)
- *2016 IEEE International Conference on Tools with Artificial Intelligence (IEEE ICTAI 2016)*; (3 times)
- *2015 IEEE International Conference on Bioinformatics and Biomedicine (IEEE BIBM 2015)–Workshop on Semantic Data Analytics and Bioinformatics (SDAB)*; (2 times)
- Registration coordinator for *The 8-th International Symposium on Chinese Spoken Language Processing (ISCSLP) 2012*
- Registration coordinator for *The 19-th International Conference on Digital Signal Processing (DSP) 2014*

MEMBERSHIP

- **Senior Member**, Institute of Electrical and Electronics Engineers (IEEE)
- **Member**, American Association of Cancer Research (AACR)
- **Member**, American Medical Informatics Association (AMIA)
- **Member**, International Society for Computational Biology (ISCB)
- **Member**, Association for Computing Machinery (ACM)
- **Member**, Organization for Human Brain Mapping (OHBM)
- **Member**, Fred and Pamela Buffet Cancer Center (FPBCC) at UNMC
- **Member**, Children Health Research Institute (CHRI) at UNMC
- **Member**, Alcohol Center of Research: Nebraska (ACORN) at UNMC

Curriculum Vitae

- **Member**, Nebraska Center for Women's Health Research (NCWHR) at UNMC
- **Member**, Center for Heart and Vascular Research (CHVR) at UNMC